

• The probability Distribution

↳ A mathematical function that maps out all possible values a variable can take alongside the actual likelihood or chance of each value occurring.

↳ Everything that can happen (possible outcomes)

↳ The exact chance of each thing happening (the possibilities)

• Sum of probability distribution = 1

Example : $P(\text{Season} | \text{Weather})$

Season	Weather					
	Sun	Rain	Cloud	Snow		
Spring	0.07	0.03	0.10	0.06	→	0.26 $P(\text{Spring})$
Summer	0.13	0.01	0.05	0.01	→	0.20 $P(\text{Summer})$
Autumn	0.05	0.05	0.15	0.03	→	0.28 $P(\text{Autumn})$
Winter	0.05	0.01	0.10	0.10	→	0.26 $P(\text{Winter})$
	↓ 0.30	↓ 0.10	↓ 0.40	↓ 0.20		

Sum of Probability distribution →

$$P(\text{Spring}) + P(\text{Summer}) + P(\text{Autumn}) + P(\text{Winter})$$

$$= 0.26 + 0.20 + 0.28 + 0.26 = 1$$

$$P(\text{Weather} = \text{Sun} \cap \text{Season} = \text{Summer}) = 0.13$$

Conditional Probability

↳ Posterior probability

↳ Notation: $P(A|B) \Rightarrow$ Probability of A given that we know B

Example: $P(\text{Sunny} | \text{Summer}) = 0.65 \Rightarrow$ Summer season; so weather sunny $\Rightarrow$ probability

$$\rightarrow \frac{P(\text{Sunny} \cap \text{Summer})}{P(\text{Summer})} = \frac{0.12}{0.20}$$

↳ Defines as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{if } P(B) \neq 0$$

↳ Can be rewritten as the product rule:

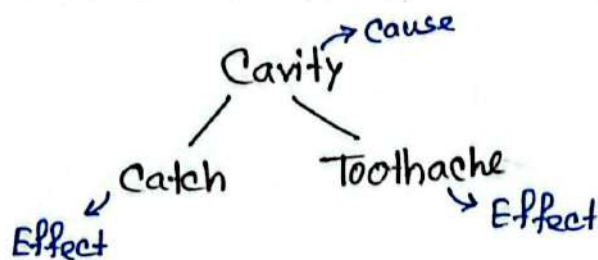
$$P(A \cap B) = P(A|B) P(B)$$

$$P(B \cap A) = P(B|A) P(A)$$

} same

For A and B to be true $\rightarrow$ B has to be true
 $\rightarrow$ A has to be true

Inference by Enumeration



	toothache		¬toothache	
	catch	¬catch	catch	¬catch
cavity	·108	·012	·072	·008
¬cavity	·16	·064	·144	·576

$$P(\text{toothache}) = 0.108 + 0.012 + 0.016 + 0.064$$

$$= 0.2$$

CAT
CATT
TCAT

$$P(\text{cavity} \vee \text{toothache}) = 0.108 + 0.012 + 0.072 + 0.008$$

$$+ 0.016 = 0.28$$

	toothache		¬toothache	
	catch	¬catch	catch	¬catch
cavity	·108	·012	·072	·008
¬cavity	·16	·64	·144	·576

Conditional probability:

$$P(\neg \text{cavity} | \text{toothache}) = \frac{P(\neg \text{cavity} \wedge \text{toothache})}{P(\text{toothache})}$$

$\nearrow \frac{P(\neg C | T)}{P(T)}$
 $= \frac{P(\neg C \wedge T)}{P(T)}$

$$= \frac{0.016 + 0.0064}{0.108 + 0.012 + 0.016 + 0.064}$$

$$= 0.4$$

Q. $P(c \wedge T \wedge k)$

$P(T|c)$

$P(c \wedge T \wedge k)$

$c = \text{cavity}$

$T = \text{toothache}$

$k = \text{catch}$

$$P(c \wedge T \wedge k) = 0.108$$

$$P(c \wedge \neg T \wedge k) = 0.72$$

$$= 0.144$$

$$P(T|c) = \frac{P(T \wedge c)}{P(c)}$$

$$= \frac{0.108 + 0.12}{0.108 + 0.012 + 0.072 + 0.008} = 0.026\%$$

□ Normalization

↳ Taking a list of row scores and changing them so they add up to exactly 1 (or 100%)

↳ Turns random point into real percentage.

	toothache		$\neg$ toothache	
	catch	$\neg$ catch	catch	$\neg$ catch
cavity	0.108	0.012	0.072	0.008
$\neg$ cavity	0.016	0.064	0.144	0.578

* Denominator can be viewed as a normalization constant α .

$$P(\text{Cavity} | \text{toothache}) = \alpha P(\text{Cavity}, \text{toothache})$$

$$= \alpha [P(\text{Cavity}, \text{toothache}, \text{catch}) + P(\text{Cavity}, \text{toothache}, \neg \text{catch})]$$

$$= \alpha [(0.108, 0.016) + (0.012, 0.064)]$$

$$= \alpha (0.012, 0.08) = (0.6, 0.4)$$

$$\begin{aligned} P(\text{C}|\text{T}) &= \frac{P(\text{C}, \text{T})}{P(\text{T})} \\ &= \frac{0.108 + 0.012}{0.20} \\ &= \frac{0.120}{0.20} = 0.6 \end{aligned}$$

$$\begin{aligned} P(\text{C}|\neg \text{T}) &= \frac{P(\text{C}, \neg \text{T})}{P(\neg \text{T})} \\ &= \frac{0.016 + 0.064}{0.80} \\ &= \frac{0.080}{0.80} = 0.4 \end{aligned}$$

$$\frac{1}{P(\text{T})} \propto$$

• General Idea $\Rightarrow$ Compute distribution on query variable by fixing

evidence variable and summing over hidden variables.

$\swarrow$
clues we can see

$\searrow$
what we want to discover

□ Independence

$\hookrightarrow$ A relation where knowing the outcome of one event provides absolutely no information about the other.

$\hookrightarrow$ Do not affect each other

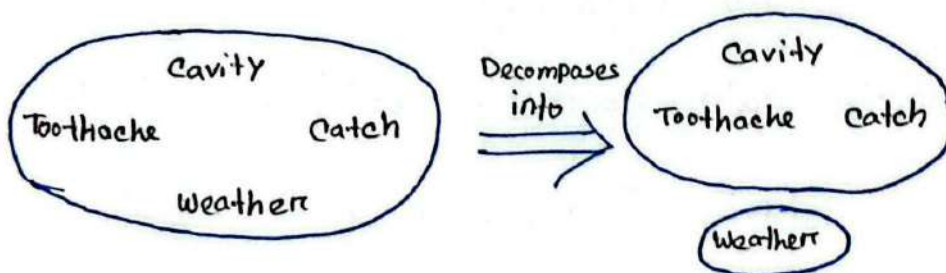
$\hookrightarrow$ Knowing A tells you nothing about B

$\hookrightarrow$ A and B are independent if -

$P(A|B) = P(A)$ or $\rightarrow$ knowing B happened doesn't change the probability of A

$P(B|A) = P(B)$ or $\rightarrow$ knowing A happened doesn't change the probability of B

$P(A, B) = P(A)P(B) \Rightarrow P(A \cap B) = P(A)P(B)$ $\rightarrow$ The joint probability of both happening together is just their individual probabilities multiplied.



$P(\text{Toothache}, \text{Catch}, \text{Cavity}, \text{Weather})$

$$= P(\text{Toothache}, \text{Catch}, \text{Cavity}) P(\text{Weather})$$

↳ For n independent biased coins $2^n \rightarrow n$.

Conditional Independence

↳ Two things might seem connected or dependent at first glance, but they are actually independent given a third factor

↳ After knowing about the third factor, the connection between first two disappears. ↳ Hidden variable

Example:

Toothache and catch → Generally happens together, so they look dependent

→ Conditionally independent given the hidden variable Cavity.

** If it is known that the patient has cavity, knowing about toothache doesn't change the probability of catch

↳ Both are separate independent symptoms of the same root cause.

• Joint distribution using chain rule-

$$P(\text{Toothache}, \text{Catch}, \text{Cavity})$$

$$= P(\text{Toothache} | \text{Catch}, \text{Cavity}) P(\text{Catch}, \text{Cavity})$$

$$= P(\text{Toothache} | \text{Catch}, \text{Cavity}) P(\text{Catch} | \text{Cavity}) P(\text{Cavity})$$

$$= P(\text{Toothache} | \text{Cavity}) P(\text{Catch} | \text{Cavity}) P(\text{Cavity})$$

$$\left| \begin{array}{l} P(T \wedge K \wedge C) \\ P(A|B) = \frac{P(A \wedge B)}{P(B)} \\ \Rightarrow P(A \wedge B) = P(A|B) P(B) \end{array} \right.$$



- In most cases, the use of conditional independence reduces the size of the representation of the joint distribution from exponential n to linear in n .

- Conditional independence is our most basic and robust form of knowledge about uncertain environments.